Week 13 Self-Assessment

Confidence Intervals, Bootstrap, and Effect Sizes

Gordon Wright

1 Instructions

Test your understanding of confidence intervals, bootstrap methods, and effect sizes. These concepts are fundamental to understanding uncertainty in our estimates and the practical significance of research findings.

---

2 Part 1: Conceptual Questions

2.1 Question 1: What is a Confidence Interval?

A 95% confidence interval tells us:

- (A) There is a 95% chance the true population value is within this range
- (B) If we repeated this study many times, 95% of the CIs would contain the true population value
- (C) 95% of our sample data falls within this range
- (D) We are 95% sure our hypothesis is correct

2.2 Question 2: Interpreting CI Width

A wider confidence interval indicates:

- (A) Greater precision in our estimate
- (B) Greater uncertainty in our estimate
- (C) A larger effect size
- (D) A significant result

What factors can lead to a narrower (more precise) confidence interval?

- (A) Larger sample size and smaller variability
- (B) Smaller sample size and larger variability
- (C) Higher confidence level (e.g., 99% instead of 95%)
- (D) More extreme scores in the data

2.3 Question 3: CI and Statistical Significance

If a 95% CI for the difference between two group means does NOT include zero, this suggests:

- (A) The result is not significant
- (B) The result is statistically significant at $p < .05$
- (C) The effect size is large
- (D) We need a larger sample

If a 95% CI for a correlation coefficient includes zero, this suggests:

- (A) The correlation is not statistically significant at $p < .05$
- (B) The correlation is significant
- (C) We should use a different test
- (D) The sample is too large

2.4 Question 4: Bootstrap Methods

The bootstrap method involves:

- (A) Collecting new participants
- (B) Using a larger sample
- (C) Resampling with replacement from the original data to estimate sampling variability
- (D) Removing outliers from the data

Why might we use bootstrap methods instead of traditional formulas?

- (A) Bootstrap is always more accurate
- (B) Bootstrap makes fewer assumptions about the data distribution
- (C) Bootstrap is required for all modern statistics
- (D) Bootstrap gives smaller p-values

2.5 Question 5: Bootstrap Confidence Intervals

A bootstrap 95% CI is constructed by:

- (A) Adding and subtracting 2 standard errors from the mean
- (B) Finding the 2.5th and 97.5th percentiles of the bootstrap distribution
- (C) Using the normal distribution tables
- (D) Calculating the exact probability

How many bootstrap samples are typically used?

- (A) 10
- (B) 100
- (C) 1000 or more
- (D) Exactly 95

2.6 Question 6: Effect Sizes

An effect size tells us:

- (A) Whether the result is statistically significant
- (B) The magnitude or practical importance of an effect
- (C) The probability of replication

- (D) The sample size needed

Why report effect sizes alongside p-values?

- (A) Because journals require it
- (B) Because statistical significance doesn't tell us about practical importance
- (C) Because effect sizes are always larger than p-values
- (D) Because p-values are unreliable

---

3 Part 2: Interpretation Questions

3.1 Question 7: Interpreting a CI

A study reports: "Mean reaction time = 450ms, 95% CI [420, 480]"

The point estimate is:

- (A) 420ms
- (B) 450ms
- (C) 480ms
- (D) 60ms

The margin of error is approximately:

- (A) 420ms
- (B) 450ms
- (C) 30ms
- (D) 480ms

We can be reasonably confident the true population mean is between:

- (A) 420ms and 480ms
- (B) 0ms and 450ms
- (C) 450ms and 900ms
- (D) We cannot say

3.2 Question 8: Comparing Groups with CIs

Study A finds: Group 1 mean = 50, 95% CI [45, 55]; Group 2 mean = 60, 95% CI [55, 65]

The CIs do NOT overlap. This suggests the difference is:

- (A) Not significant
- (B) Likely statistically significant
- (C) Due to chance
- (D) Meaningless

Study B finds: Group 1 mean = 50, 95% CI [40, 60]; Group 2 mean = 55, 95% CI [45, 65]

The CIs DO overlap substantially. This suggests:

- (A) The groups are definitely different
- (B) The difference may not be statistically significant
- (C) We should increase the confidence level
- (D) The study failed

3.3 Question 9: Cohen's d Interpretation

Cohen's d = 0.20 is considered:

- (A) Small
- (B) Medium
- (C) Large
- (D) Negligible

Cohen's d = 0.50 is considered:

- (A) Small
- (B) Medium
- (C) Large
- (D) Very large

Cohen's d = 0.80 is considered:

- (A) Small
- (B) Medium
- (C) Large
- (D) Massive

Cohen's d = 1.20 would be considered:

- (A) Small
- (B) Medium
- (C) Large (or very large)
- (D) Impossible

3.4 Question 10: Interpreting Effect Size in Context

A medication reduces anxiety scores with d = 0.25 (small effect). A therapy reduces anxiety with d = 0.75 (medium-large effect).

Which treatment shows a larger practical effect?

- (A) Medication
- (B) Therapy
- (C) They are the same
- (D) Cannot compare

If the medication is much cheaper and has no side effects, should we ignore it because the effect is "small"?

- (A) Yes, small effects are worthless

- (B) No, small effects can still be meaningful in context
- (C) Yes, only large effects matter
- (D) We should only consider statistical significance

---

4 Part 3: Application Questions

4.1 Question 11: Scenario - Understanding Uncertainty

A researcher measures spelling test scores in 30 children and reports: M = 75, 95% CI [70, 80].

Which statement is the best interpretation?

```
opts_ci <- c(
  "95% of children scored between 70 and 80",
  "There is a 95% probability the true mean is between 70 and 80",
  answer = "We are 95% confident that the interval from 70 to 80 captures the
true population mean",
  "The mean could be any value from 0 to 100"
)
```

- (A) 95% of children scored between 70 and 80
- (B) There is a 95% probability the true mean is between 70 and 80
- (C) We are 95% confident that the interval from 70 to 80 captures the true population mean
- (D) The mean could be any value from 0 to 100

4.2 Question 12: Scenario - Choosing Methods

A researcher has data that is heavily skewed (not normally distributed). They want to estimate the confidence interval for the median.

Which approach is most appropriate?

- (A) Use the standard formula assuming normality
- (B) Use bootstrap methods
- (C) Ignore the skewness
- (D) Only report the mean

4.3 Question 13: Scenario - Practical Significance

A pharmaceutical company tests a new drug for depression. They find a statistically significant improvement ($p = .02$) with $d = 0.15$.

How should they interpret this?

```
opts_drug <- c(
  "The drug is highly effective and should be approved immediately",
```

```
  "The result is not significant, so the drug doesn't work",
  answer = "The effect is statistically significant but very small - clinical
utility is questionable",
  "Effect sizes don't matter for drug trials"
)
```

- (A) The drug is highly effective and should be approved immediately
- (B) The result is not significant, so the drug doesn't work
- (C) The effect is statistically significant but very small - clinical utility is questionable
- (D) Effect sizes don't matter for drug trials

---

5 Part 4: MCQ Exam Practice

5.1 Investigation 1

A researcher measures memory scores in a sample of 40 participants and calculates: M = 65, SD = 12, 95% CI [61.17, 68.83].

1. **What does the CI tell us about precision?**

- (A) Very precise (narrow interval)
- (B) Moderately precise
- (C) Very imprecise (wide interval)
- (D) Cannot determine

2. **If they had tested 100 participants instead, the CI would likely be:**

- (A) Wider
- (B) Narrower
- (C) The same
- (D) Impossible to predict

3. **If the population mean is actually 64, would this CI capture it?** TRUE / FALSE

4. **The margin of error is approximately:**

- (A) 65
- (B) 12
- (C) About 4
- (D) 61

5.2 Investigation 2

A study compares two teaching methods. Results show: Method A (M = 78, SD = 10) vs Method B (M = 72, SD = 11), with Cohen's d = 0.57 and p = .03.

1. **The effect size is:**

- (A) Small
- (B) Medium

- (C) Large
- (D) Not significant

2. **Is the result statistically significant at alpha = .05?** TRUE / FALSE
3. **Method A shows scores that are on average how much higher?**

- (A) 0.57 points higher
- (B) 6 points higher
- (C) 57% higher
- (D) Cannot calculate

4. **The d = 0.57 means the difference is about:**

- (A) 0.57 points
- (B) About half a standard deviation
- (C) 57 standard deviations
- (D) Not meaningful

5. **Should we conclude Method A is practically better?**

- (A) No, the effect is too small
- (B) Possibly - a medium effect in education can be meaningful
- (C) Yes, definitely superior
- (D) Cannot say without more data

---

6 Summary Checklist

Before moving on, make sure you can:

- ☐ Explain what a confidence interval represents
- ☐ Interpret the width of a confidence interval
- ☐ Understand how sample size affects CI width
- ☐ Explain the basic logic of bootstrap methods
- ☐ Interpret Cohen's d values (small, medium, large)
- ☐ Distinguish statistical significance from practical significance
- ☐ Explain why effect sizes are important to report

! Key Takeaway

Statistical significance (p-value) tells us whether an effect is likely real, while effect size tells us whether it matters practically. Always consider both when interpreting research findings.